Docker-01: Docker Installation and Container Management on AWS EC2

1. Objective

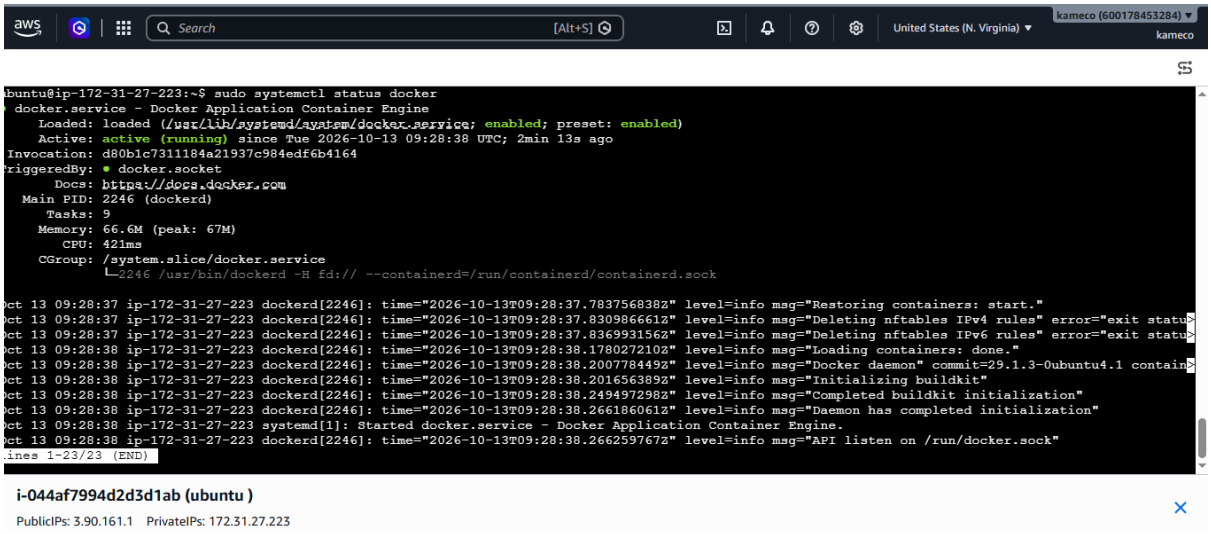
The objective of this task is to set up Docker on an Ubuntu EC2 instance and learn how to manage Docker images and containers. As part of this task, I installed Docker, downloaded five Docker images, deployed an Nginx container on port 81, removed the Apache Tomcat image, inspected the Jenkins and SonarQube images, and ran Jenkins in a container to execute a sample job.

Environment and Prerequisites

- Cloud Platform: Amazon Web Services (AWS)
- Virtual Machine: EC2 instance
- Operating System: Ubuntu Server
- Container Platform: Docker
- Web Server: Nginx
- CI/CD Tool: Jenkins
- Code Quality Tool: SonarQube
- Application Server Image: Apache Tomcat

Install Docker on Ubuntu

Objective: To install Docker so that containers and images can be managed on the EC2 instance



Pull Five Docker Images

Objective: To download the required Docker images from Docker Hub for learning image management and container execution.

Steps performed:

1. Pulled the Nginx image.
2. Pulled the Apache Tomcat image.
3. Pulled the Ubuntu image.

4. Pulled the Jenkins LTS image.
5. Pulled the SonarQube Community image.

Docker images



Nginx Container on Port 81

Objective: To run an Nginx web server inside a Docker container and make it accessible through port 81

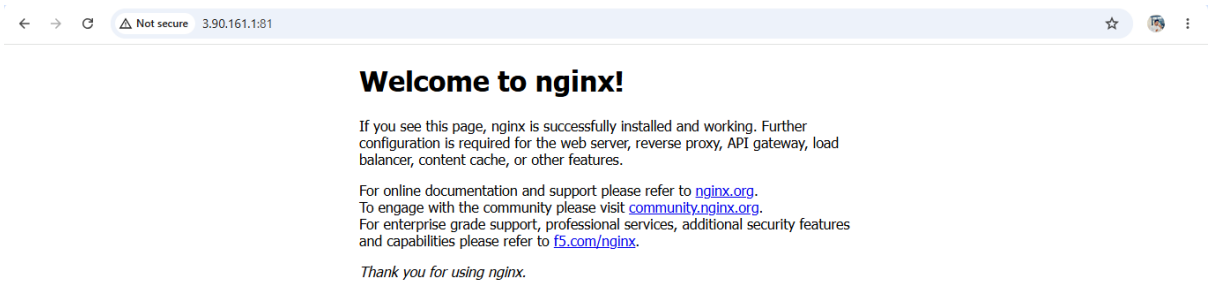
Command used:

```
sudo docker run -d --name nginx-container -p 81:80 nginx
```

Testing commands:

```
sudo docker ps
curl http://localhost:81
```

Result: The Nginx container ran successfully and served the default welcome page through port 81.



Delete the Apache Tomcat Image

Objective: To learn how to remove an unused Docker image from the local image repository.

```
ubuntu@ip-172-31-27-223:~$ sudo docker rmi tomcat
Untagged: tomcat:latest
Deleted: sha256:6974cddd46f09a10a0d4ffae6cd77f9122b3cf7de8f4b7c3305f35026accc81f
ubuntu@ip-172-31-27-223:~$ sudo docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     5e2309035332       25.9kB         9.49kB    U
jenkins/jenkins:lts    a660310e39ad        720MB          246MB
nginx:latest           f9ea18bfa4fa        243MB         66.7MB    U
sonarqube:community    2e378f7852b6       2.56GB          1.1GB
ubuntu:latest          f144425ff09b        162MB         45.6MB
ubuntu@ip-172-31-27-223:~$
```

Commands used:

```
sudo docker ps -a
sudo docker rmi tomcat
sudo docker images
```

Result: The Apache Tomcat image was removed successfully from the local Docker image repository

Run Jenkins in a Docker Container

Objective: To deploy Jenkins using Docker and access its web interface for CI/CD job execution

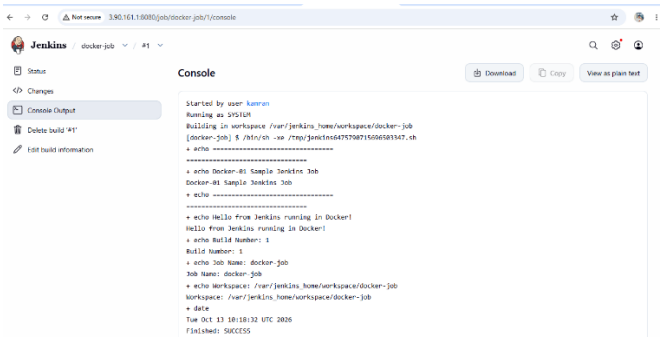
```
ubuntu@ip-172-31-27-223:~$ sudo docker images jenkins/jenkins:lts sonarqube:community
docker: 'docker images' requires at most 1 argument

Usage:  docker images [OPTIONS] [REPOSITORY[:TAG]]

Run 'docker images --help' for more information
ubuntu@ip-172-31-27-223:~$
sudo docker run -d \
  --name jenkins-container \
  --restart unless-stopped \
  -p 8080:8080 \
  -p 50000:50000 \
  -v jenkins_home:/var/jenkins_home \
  jenkins/jenkins:lts
677c2edd2f157fc5177cfb1f2a826763ec21bb100ecf357a3382e2e4f8cca48a
ubuntu@ip-172-31-27-223:~$
```

Command used:

```
sudo docker run -d \
  --name jenkins-container \
  --restart unless-stopped \
  -p 8080:8080 \
  -p 50000:50000 \
  -v jenkins_home:/var/jenkins_home \
  jenkins/jenkins:lts
```



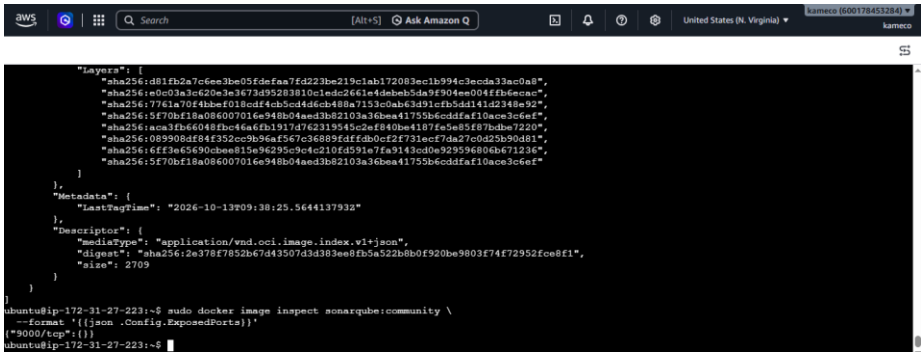
Inspect the SonarQube Image

Objective: To examine the SonarQube Docker image and understand its configuration.

Steps performed:

1. Selected the SonarQube Community image.
2. Used the Docker image inspection command.
3. Reviewed the image ID and configuration details.
4. Checked the exposed ports and configured environment variables.






```
sudo docker image inspect sonarqube:community
sudo docker image inspect sonarqube:community \
  --format '{{json .Config.ExposedPorts}}'
```

Result: The SonarQube image metadata and configuration were inspected successfully.